

SECTION 15290-A DUCT INSULATION

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of Contract, including General and Supplementary Conditions and Division 1 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section includes insulation materials and thickness for ducts, air handling units, terminal units, and exhaust mufflers.

1.3 REFERENCES

- A. Commercial and industrial insulation standards, published by Western Insulation Contractors Association (WICA, latest edition).
- B. Taiwan Energy Codes
- C. American Society for Testing and Material (ASTM).
- D. Factory Mutual (FM).

1.4 SUBMITTALS

- A. Submit shop drawings and samples in accordance with Division 1.
- B. One sample each of the type of pipe cover insulation to be used.
- C. Manufacturer's data sheets for each type of insulation or accessory material to be used.
- D. Details of pipes through fire-rated walls/roof indicating FM approval and the requirements thereof.
- E. Submit manufacturer's installation instructions.

1.5 PRODUCT DELIVERY, INSPECTION, AND STORAGE

- A. Delivery:

1. All materials shall be delivered in the manufacturer's original unopened protective package.
 - B. Receiving Inspection:
 1. Materials shall be inspected by the Contractor prior to acceptance.
 - C. Storage and Handling:
 1. Materials shall be resealed or repackaged after inspection and stored in the original protective packaging or approved equal.
- 1.6 DEFINITIONS
- A. Exposed defines ducts which are visible, as in equipment rooms, on the roof, in service tunnels, and in rooms without ceilings.
 - B. Concealed defines ducts which are not normally visible, as in plenums, chases, shafts, and above ceilings.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Adhesives and Insulation Materials: Composite fire and smoke hazard rating maximum 25 for flame spread and 50 for smoke developed per ASTM E 84. Adhesives shall be waterproof and meet ASTM C 916 and FM approve.
- B. Identification: Provide composite material with legibly printed manufacturer's name, nominal thickness, flame spread, and smoke developed ratings per NFPA 90A and UL 181.

2.2 MATERIALS

- A. Exposed HVAC Ducts: Cover with rigid glass fiber insulation board (ASTM C 612) with K-value at 75 degrees F, maximum 0.24 Btu-in/hr-sq ft degrees F, 6 pcf minimum density, with factory-applied, glass-reinforced kraft paper all service vapor barrier jacket. Maximum permeability of vapor barrier to be 0.02 perms. Finish with aluminum cladding jacket.
- i) Supply air ducts insulation thickness=1" (FM)
 - ii) Return air ducts insulation thickness=1"
 - iii) Outside air ducts insulation thickness=1"
- * No insulation on MAU makeupair (Outside air) ducts *
- B. Outdoor HVAC Ducts: Same as exposed ducts with a weather resistant aluminum sheet metal cladding jacket. Seams shall be caulked for water tight cover; Ducts with a weather resistant SUS304 sheet metal come with powder (烤漆) baked coating cladding jacket. Seams (風管法蘭處) shall be caulked for water tight cover.
- Air Handlers (Ducts Outside of Building)
- i) Supply air ducts = 2"
 - ii) Return air ducts = 2"
 - iii) Outside air ducts = no insulation
 - iv) Outside air ducts within conditioned space = 2"
 - iv) Inlet ducts = 2"
 - v) Outlet ducts = 2"
- C. Concealed HVAC Ducts: Cover with flexible glass fiber insulation, ASTM C 1290, K-value at 75 degrees F, maximum 0.29 Btu-in/hr-sq ft degrees F, 3/4 pcf minimum density, with factory-applied, glass-reinforced aluminum foil and kraft paper vapor barrier jacket. Maximum permeability of vapor barrier to be 0.02 perms.
- i) Supply air ducts insulation thickness=1 inch
 - ii) Return air ducts insulation thickness=1 inch
- D. Scrubbed Exhaust Ducts:
- flexible, black, closed cell foam insulation with K-value at 75 degrees F maximum 0.27 Btu-in/hr-sq ft degrees F.

PART 3 - EXECUTION

3.1 PREPARATION

- A. Ensure surface is clean and dry prior to installation. Ensure insulation is dry before and during application.

3.2 INSTALLATION

- A. Ensure insulation is continuous through inside wall penetrations. Exception, fire dampers and penetrations detailed otherwise.
- B. Finish insulation neatly at hangers, supports and other protrusions. Coat with vapor barrier mastic. Refer to Sections 15245 and 15140.
- C. Where insulation requires sealing and taping, complete procedures at the same time insulating material is being applied to assure clean surfaces exist for proper adhesion. (風管施工前表面清潔)
- D. D. Locate insulation or cover seams in least visible locations. (外觀)
- E. Where insulation requires sealing and taping, complete procedures at the same time insulating material is being applied to assure clean surfaces exist for proper adhesion.
- F. Locate insulation or cover seams in least visible locations.
- G. Secure rigid insulation with a minimum 80 percent coverage of adhesive and welded cupped head pins 16 inches on center, all 4 sides of duct. Use of stickpins not permitted. All corners shall be protected with galvanized steel corner beads firmly held in place. Seal joints and breaks with 4 inch wide strips of open mesh glass cloth or tape embedded between 2 coats of vapor barrier sealant. Point up other joints and breaks with hydraulic setting cement.
- H. Cover ductwork located outdoors with rigid insulation as specified above. In addition, apply a weather barrier mastic reinforced with fabric or mesh over the exterior surface. Reference, Plate No. 20, WICA insulation standards.
- I. Secure flexible insulation to ductwork surfaces with 100 percent coverage of UL listed adhesive and welded cupped head pins, 12 inch on center, all 4 sides of duct. Use of stickpins not permitted. Butt insulation and seal joints and breaks with 2 inch lap of foil adhered over joint.

- J. Engine exhaust and Site fabricated steel breeching insulation shall be tightly butted with staggered joints secured with metal bands or wire. Cover fittings with equivalent thickness of hydraulic setting cement applied over wire mesh. Cover with 20 gauge aluminum sheet.
- K. Apply 1 layer of exposed duct insulation to terminal units, coils, duct transitions, and attenuators located downstream of units designated to have internal aluminum foil facing.

3.3 INSULATION SCOPE

System	Scope	Thickness (inches)
Air Handlers (All Ducts in Air Conditioned Spaces)	Supply air ducts	1-1/2
* No insulation on Fab makeup air ducts *	Return air ducts	1-1/2
	Outside air ducts	1-1/2
Air Handlers (Ducts Outside of Building)	Supply air ducts	2
	Return air ducts	2
	Outside air ducts	None
	Outside air ducts within conditioned space	1
Air Handlers (Ducts in non-Air Conditioned Spaces in Building)	Supply air ducts	1-1/2
	Return air ducts	1-1/2
	Outside air ducts	1-1/2
Terminal Units	Inlet ducts	1-1/2
	Outlet ducts	1 (lined)
Scrubbed Exhaust Ducts	All metal ducts downstream from the scrubber discharge, indoor ducts only	1
Exhaust	Duct within 15' of point of discharge	1

System	Scope	Thickness (inches)
Engine Exhaust and Muffler	Welded steel, inside building	2
Aluminum Foil Faced Terminal Units	Terminal Units, Coils, Duct Transition and Attenuator Cover with flexible, black, closed cell foam insulation Coils, Duct Transition and Attenuator =2 "	2

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